MASAN GROUP

MASAN GROUP Information Security Office

Benchmark AntiMalware Report

Microsoft vs Kaspersky

Statement of Confidentiality

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I. Revision History

Rev	Revision History	Date
1	Initial version of report	August 15, 2024

II. References

Ref #	Document	Author	Date
1	Kaspersky - Machine Learning in Cybersecurity	AO Kaspersky Lab.	2024
2	Microsoft Defender for Endpoint	Microsoft Corporation	2024
3	Kaspersky Endpoint Security 12.6 for Windows	AO Kaspersky Lab	2024

III. Executive Summary

3.1 Summary

According to the cybersecurity attack report that our team conducted at WinCommerce (WCM), the Trellix malware protection system currently in use by WCM, Masan Consumer Holdings (MCH), and Masan Group (MSN) is a significant vulnerability in our security infrastructure. Notably, recent ransomware attacks in Vietnam, such as the case with PV Oil, were also associated with the use of this Trellix solution.

Therefore, as per the mandatory recommendation in the report, we have selected and conducted a technical evaluation of two alternative solutions with better quality:

- Microsoft Defender for Endpoint Plan 2
- Kaspersky Next EDR Foundation

The results show that Kaspersky is the optimal choice in terms of quality, responsiveness, and effectiveness in malware prevention, outperforming Microsoft. Additionally, Kaspersky is more cost-

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effective compared to Microsoft. For further details on the evaluation results, please refer to the attached detailed report.

However, please note that switching to Kaspersky will incur additional costs. Currently, Trellix is being used beyond the licensed capacity in Vietnam, which provides favorable pricing. Therefore, to optimize the use of Kaspersky, I propose dividing the users into the following groups:

3.2 Scope

This report presents our results and findings on the comparison of performance, quality, features, and cost between Microsoft Defender for Endpoint Plan 2 (Defender) and Kaspersky Next EDR Foundations (Kaspersky). The benchmarks conducted on both products are as follows:

- **Cost**
- **Performance**
- **Features:**
    - Advanced Threat Prevention (AI, ML)
    - Always Protection (Online and Offline)
    - Firewall
    - Web Control
    - Network Protection
    - Threat Protection
    - Log Inspection
    - Web Threat Protection
    - Threat Intelligence
    - Quarantine Malicious Non-PE Files (MS-Office, PDF, ISO, Zip, 7zip, Images, Video, etc.)
    - Device Control
    - Behavioral-Based Detection of Threats
    - Signature-based Detection of Threats (Built-In NextGen AV)
    - Real-Time Monitoring System Activity
    - Sandbox Analyzer for Dynamic Analysis
    - Unusual Files and Registry Entries
    - Fileless Malware
    - Ransomware
    - File Integrity Monitoring
    - User and Entity Behavioral Analytics
    - Root Cause Analysis

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3.3 Project Team

The engagement involved contributions from the following team members:

ISO Team	Role
Nguyễn Hồng Sơn	Security Expert
Nguyễn Quốc Việt	Senior Security Engineer

3.4 Conclusion

The team has built test cases that are suitable for the environment of the Win commercial information system of Masan Group. In addition, the team also checked whether the solutions support the necessary features to serve the needs of the operations team.

The team will focus on the features and compare the quality of these features of the 2 solutions, with features that are difficult to compare, we can only check whether the solutions support them. For the features that can be evaluated for quality, we define them as follows:

Rank	Description
N/A	No threats are detected, or features are available. No warnings or preventive actions are taken, leaving the system completely vulnerable.
1 - Minimal	detects only the most basic and well-known threats. Detection is sporadic and often misses more advanced or newly emerging threats. False negatives are common, and prevention mechanisms are weak or ineffective.
2 - Basic	detects a reasonable number of common threats. It can identify some known threats but struggles with advanced or sophisticated malware. Detection rates are moderate, and while it can block certain threats, its effectiveness against zero-day attacks and advanced persistent threats (APTs) is limited.
2 - Moderate	detects a broad range of threats, including some advanced threats and variations of known malware. It performs reasonably well against common attack vectors but may still miss more sophisticated or customized attacks. False positives and negatives are relatively balanced, and prevention capabilities are generally effective.
4 - High	effectively detects and prevents a wide variety of threats, including many advanced and emerging threats. Its machine learning and behavior-based detection methods provide strong protection against zero-day vulnerabilities and APTs. False positives are minimal, and the solution is reliable for most environments.

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5 - Exceptional	offers top-tier threat detection and prevention, consistently identifying and mitigating even the most sophisticated and stealthy threats. It excels in zero-day protection, APT detection, and response, with minimal false positives and a strong, proactive defense strategy

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Below is the table of results based on the team's evaluation and the results are as follows:

Category	Description	Defender for Endpoint P2	Kaspersky EDR Foundations
PROTECTION from THREATS	Advanced Threat Prevention (AI, ML)	Support	Support
	Always Protection (Online and Offline)	Support	Support
	Firewall	Support	Support
	Web Control	Support	Support
	Network Threat Protection	N/A	Support
	Mail Threat Protection	N/A	Support
	Web Threat Protection	3	5
	Threat Intelligence	Support	Support
	Quarantine Malicious Non-PE Files (MS-Office, PDF, ISO, Zip, 7zip, Images, Video, etc.)	4	5
	Device Control	Intune	Support
DETECTION of THREATS	Behavioral-Based Detection of Threats	3	5
	Signature-based Detection of Threats (Built-In NextGen AV)	5	4
	Real-Time Monitoring System Activity	N/A	Support
	Sandbox Analyzer for Dynamic Analysis	Cloud	Local
	Unusual Files and Registry Entries	5	5
	Fileless Malware	5	5
	Ransomware	Support	Support
	Root Cause Analysis	Support	Support

Looking at the comparison table, we see the result:

- Kaspersky is more prominent in behavioral-based threat detection and prevention than Defender. *Kaspersky is better at detecting complex malware than Defender.*
- Defender is a built-in package on Windows, so it recognizes Signature-based Detection of Threats *better* than Kaspersky.

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- Kaspersky's mail protection and network protection features prevent malware from being transmitted via email, and its scanning capabilities within the internal network are extremely good, while *Defender doesn't support these features.*
- Kaspersky's Web Threat Protection feature also protects malicious websites *better* than Defender.
- Kaspersky's non-PE malicious file quarantine mechanism (MS-Office, PDF, ISO, Zip, 7zip, Images, Videos, etc.) *is better than Defender.*

With the same support features:

- Kaspersky is *more friendly and easier to use*.
- Control Device is centrally managed on the administrator dashboard for Kaspersky, but Defender must integrate many other tools such as Microsoft Intune.

With the cost and performance:

Regarding the performance evaluation of solutions affecting IT systems, due to limitations in the environment and performance measurement tools, we used reference information from a reputable site with the following statistics:

Benchmarks	Microsoft Defender	Kaspersky
One Time Service Cost	5.2 $ user/ month	~2.8$/user/ year
Slowing-down when launching popular websites	-14%	-16%
Slower download of frequently-used applications	0%	0%
Slower launch of standard software applications	-9%	**-7%**
Slower installation of frequently-used applications	-23%	**-11%**
Slower copying of files, locally and in a network	0%	1%

- The above cost cannot use all the features of Microsoft Defender, you must pay **additional costs** for the accompanying tools.
- Kaspersky's hardware usage is 2% higher than Defender's, but the quality is significantly better.

IV. Methodology and Result

Benchmark 1 – Production Metric

In addition to the evaluation criteria for the quality of attack detection and prevention, the team also collects statistics and has a comparison table of criteria such as supported operating systems, resource requirements or the ability to combine with other products, specifically as follows:

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Category	Description	Defender for Business	Kaspersky EDR Foundations
Product Adaptability	Multi-Platform Support both IT and OT (Embedded OS, Win7, Win10, Win11, Linux, MacOS, iOS, Android)	**Windows:** *Windows 11 Enterprise / IoT Enterprise / Education/ Pro / Pro Education* *Windows 10 Enterprise / Enterprise LTSC 2016 (or later) / IoT Enterprise (including LTSC) / Education / Pro / Pro Education* *Windows server (version 1803 or later)/ Server 2012 R2 / Server 2016 / Server 2019 / Server 2019 core edition / Server 2022 / 2022 core edition* **Require the use of the Log Analytics / Microsoft Monitoring Agent (MMA):** *Windows 8.1 Enterprise / Pro* *Windows 7 SP1 Enterprise / SP1 Pro* *Windows Server 2008 R2 SP1* **macOS:** *14 (Sonoma), 13 (Ventura), 12 (Monterey)* **Linux server (AMD64/EM64T/x86_64):** *Red Hat Enterprise Linux 6.7 or higher (In preview) / 7.2 or higher / 8.x / 9.x* *CentOS 6.7 or higher (In preview)* *Ubuntu 16.04 LTS / 18.04 LTS / 20.04 LTS / 22.04 LTS* *Debian 9 - 12* *SUSE Linux Enterprise Server 12 or higher* *Oracle Linux 7.2 or higher* *Amazon Linux 2 / 2023* *Fedora 33-38*	**Windows:** *Windows 7 Home / Professional / Ultimate / Enterprise Service Pack 1 or later;* *Windows 8 Professional / Enterprise;* *Windows 8.1 Professional / Enterprise;* *Windows 10 Home / Pro / Pro for Workstations / Education / Enterprise / Enterprise multi-session;* *Windows 11 Home / Pro / Pro for Workstations / Education / Enterprise.* *Windows MultiPoint Server 2011 (64-bit);* *Windows Server 2008 R2 Foundation / Standard / Enterprise / Datacenter Service Pack 1 or later;* *Windows Web Server 2008 R2 Service Pack 1 or later;* *Windows Server 2012 Foundation / Essentials / Standard / Datacenter (including Core Mode);* *Windows Server 2012 R2 Foundation / Essentials / Standard / Datacenter (including Core Mode);* *Windows Server 2016 Essentials / Standard / Datacenter (including Core Mode);* *Windows Server 2019 Essentials / Standard / Datacenter (including Core Mode);* *Windows Server 2022 Standard / Datacenter / Datacenter: Azure Edition (including Core Mode).* **Linux 32/64 bit:** *Ubuntu 16.04 LTS* *Red Hat Enterprise Linux 6.7 / 7.2 and later* *CentOS 6.7 and later*

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		Rocky 8.7 and higher *Alma 8.4 and higher* *Mariner 2* **Android:** *Mobile phones and tablets running Android 8.0 and above. Mobile phones running Android go and other mobile devices running Android are not currently supported.* **IOS:** *iOS device running iOS 15.0 and above. iPads are also supported.*	*Debian GNU/Linux 8.6/9.4 and later* *Linux Mint 18.2/19 and later* *Alt Linux SPT 7.0.6 (work by using the graphical user interface (GUI) is not supported)* *Alt Linux SPT 8.0.0 Workstation* *Alt Linux SPT 8.0.0 Server* *Alt Linux 8.2 Workstation / Workstation K / Server / Education* *Lotos* *GosLinux 6.6* **Linux 64bit:** *CentOS 7.2 and later* *Oracle Linux 7.3 and later* *SUSE Linux Enterprise Server 15* *openSUSE 15* *Amazon Linux AMI* *Micro Focus Open Enterprise Server 2018* *Astra Linux Special Edition RUSB.10015-01 (update 1.5 and 1.6)* *Astra Linux Special Edition RUSB.10015-16 (release 1) (update 1.6)* *Zircon-36KT* *Zircon-36CT* *OS ROSA Cobalt 7.3 for client and server systems* *EMIAS 1.0* *Red OS 7.2* *Perl interpreter: version 5.10 or higher* **macOS:** *14 (Sonoma), 13 (Ventura), 12 (Monterey)* **Android**: *Android 5.0–13 (including Android 12L, excluding Go Edition)* **IOS**: *iOS 10–16 or iPadOS 13–15*
	Hardware and software requirements	**Windows**: Cores: 2 minimum, 4 preferred	**Windows**: 2 GB of free disk space on the hard drive;

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		Memory: 1 GB minimum, 4 preferred **Linux**: Disk space: 2 GB Cores: 2 minimum, 4 preferred Memory: 1 GB minimum, 4 preferred **Macos**: Supported processors: x64 and ARM64. Disk space: 1GB	CPU: Workstation: 1 GHz; Server: 1.4 GHz; Support for the SSE2 instruction set. RAM: Workstation (x86): 1 GB; Workstation (x64): 2 GB; Server: 2 GB; **Linux**: Core 2 Duo 1.86 GHz or faster processor swap partition at least 1 GB 1 GB of RAM for 32-bit operating systems, 2 GB of RAM for 64-bit operating systems 4 GB of free hard disk space for installation of the application and storage of temporary and log files **MacOS**: Processor type: Intel, Apple silicon 4 GB of memory (RAM) 5 GB of free disk space Operating system macOS 12-14
	Multi-Integration Support and Able to Co-Exist with Other Security Solution (AV, EPP, DLP, etc.)	Optimized for integration with other Microsoft products and on the Azure cloud platform.	It can be used solely for EDR features and combined with other antivirus services such as Defender.
Recover from Incidents	Run On-Demand AV	Support	Support
	Root Cause Analysis	Support	Support
Support on standard subscription	Built-In Automated Response with Playbook	Support	Support
	Built-In Vulnerability Management	No	Support
	Built-In Patch Management	N/A	Support
	24x7 On Call Support	N/A	No
	24x7 On Email Support	Support	Support

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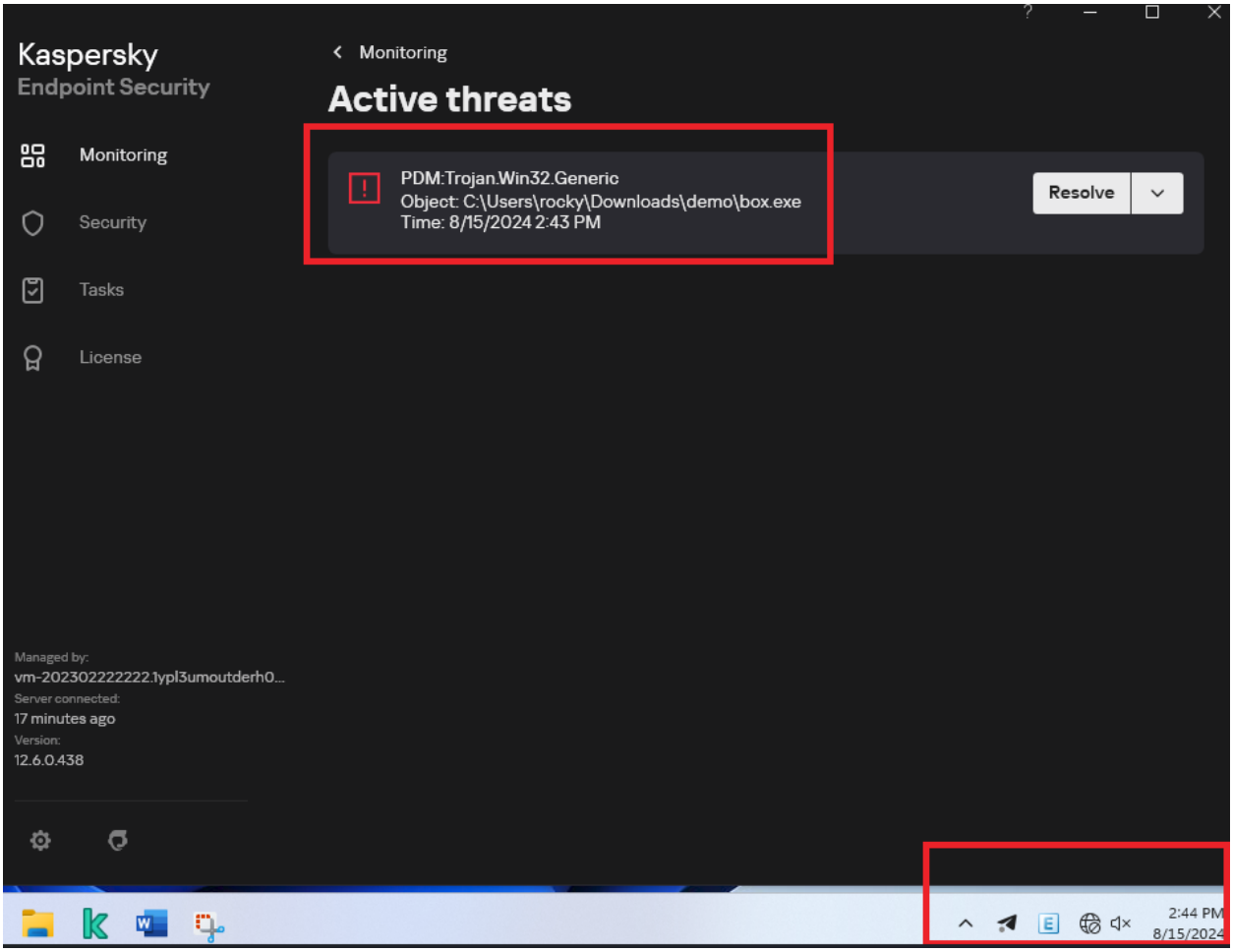
Benchmark 2 – Always Protection (Online and Offline)

The Always Protection feature of an Endpoint Detection and Response (EDR) system is important because it ensures that your security systems operate effectively both when online and offline.

Test case: We dropped malware on devices with Microsoft Defender for Enterprise and Kaspersky EDR Foundations installed in both online and offline scenarios.

Two solutions work well, with all the necessary features to detect and prevent threats in both online and offline scenarios.

Testcase	Microsoft Defender for Endpoint	Kaspersky EDR Foundations
Internet	Detected	Detected
No Internet	Detected	Detected



Kaspersky EDR detected Malware without Internet

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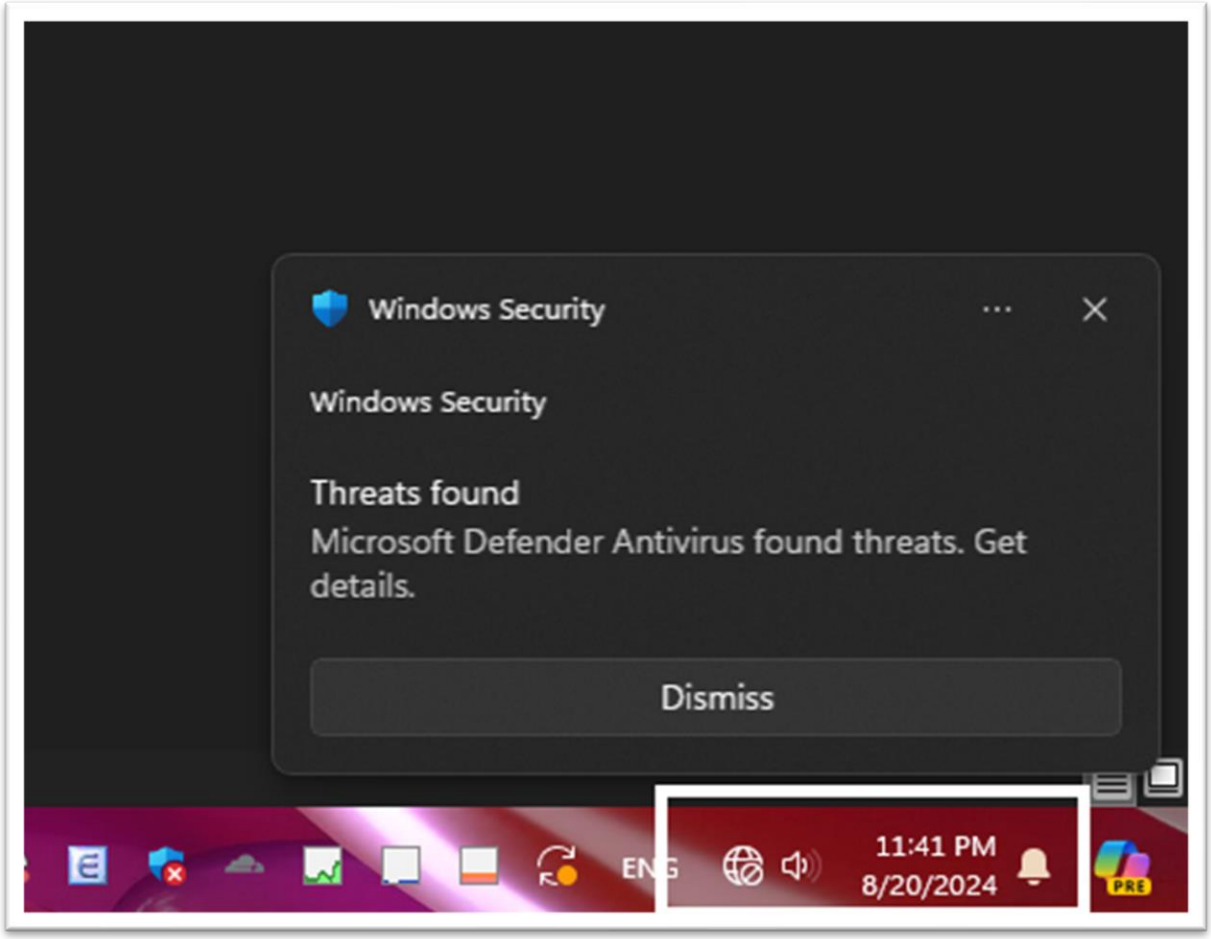

Microsoft detected Malware without Internet

Benchmark 3 – Web Control

Web control allows managing the actions of LAN users by restricting or blocking access to web resources. Web control offers the following options:

- Traffic saving
- Access differentiation based on the content category of web resources.
- Centralized control of access to web resources.

Test case: Explore and discover, experience the firewall features on both solutions. Add a rule to block content if accessing any domain.

Both solutions have the Web Control or Web Content Filter feature. This is an essential feature that helps administrators apply policies effectively to prevent risks for everyone.

Testcase	Microsoft Defender for Endpoint	Kaspersky EDR Foundations
Tested website	Blocked	Blocked

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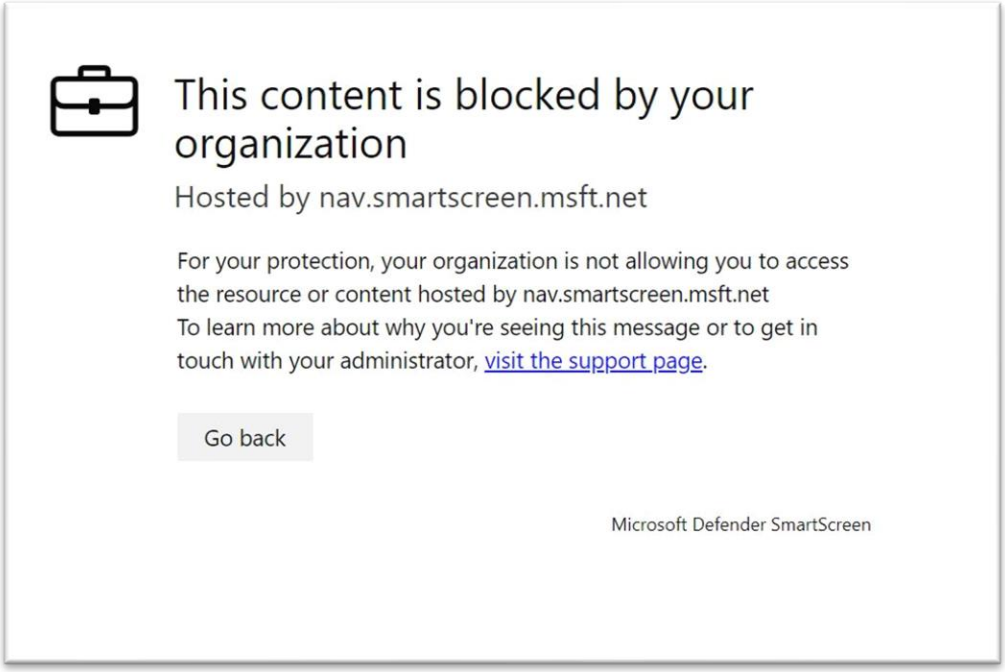
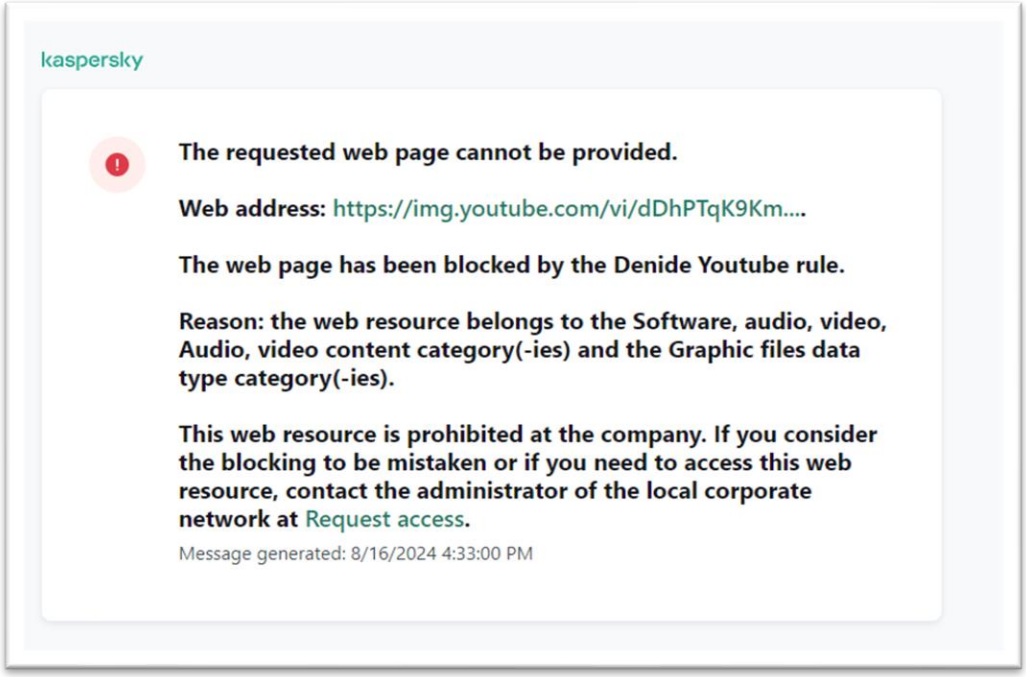


Microsoft Defender for Endpoint blocked website by content filter



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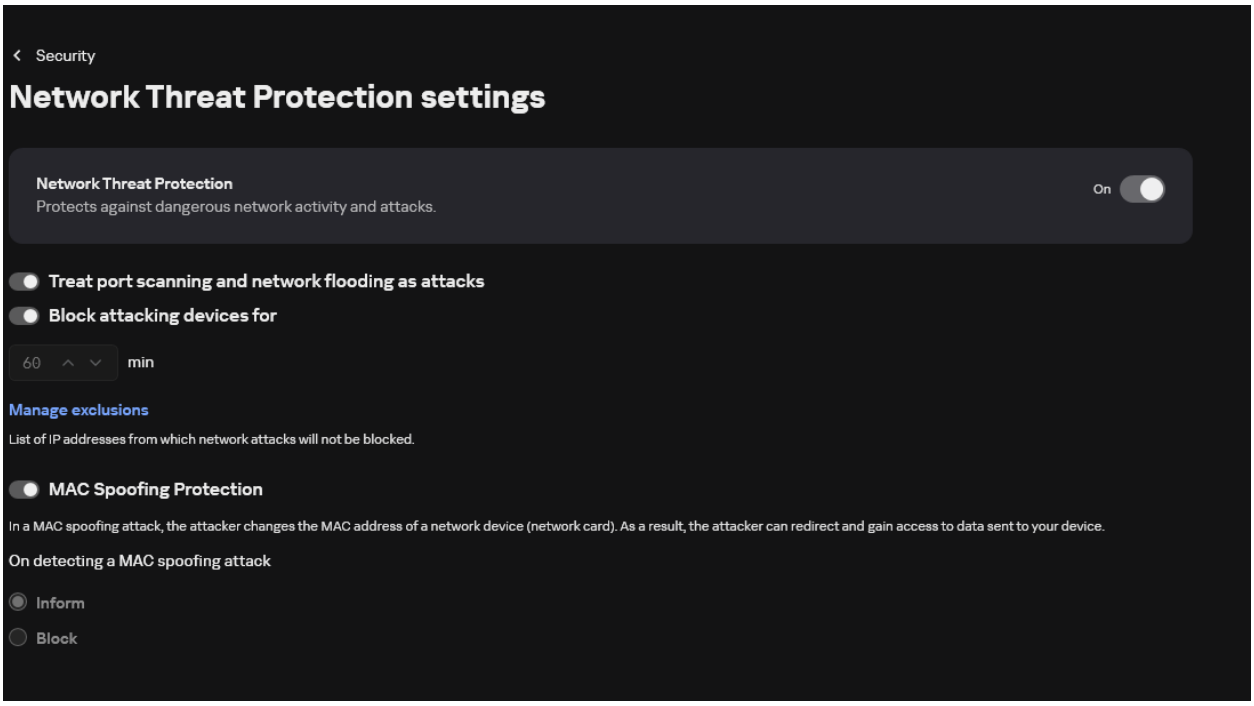
Kaspersky EDR Foundations blocked website by content filter

Benchmark 4 – Firewall and Network Theat Protection

The Network Protection component scans incoming network traffic for typical signs of cyberattacks. When a cyberattack targeting your computer is detected, the EDR solution will block network activity from the attacking computer. A warning will then appear on your screen, notifying you of the cyberattack and displaying information about the attacking computer. The **firewall** detects all network connections of the user's computer and provides a list of IP addresses, with an indicator of the status of the default network connection.

Test case: Simulate a network scanning attack using the **Nmap** tool to monitor the actions of both solutions.



Dashboard Network Threat Protection of Kaspersky

Kaspersky detected and blocked the connection from the scanning IP, but Defender did not recognize it and also did not have this feature.

Testcase	Defender	Kaspersky
Nmap Scanning	Not Blocked	Blocked
Firewall Rule	Support	Support

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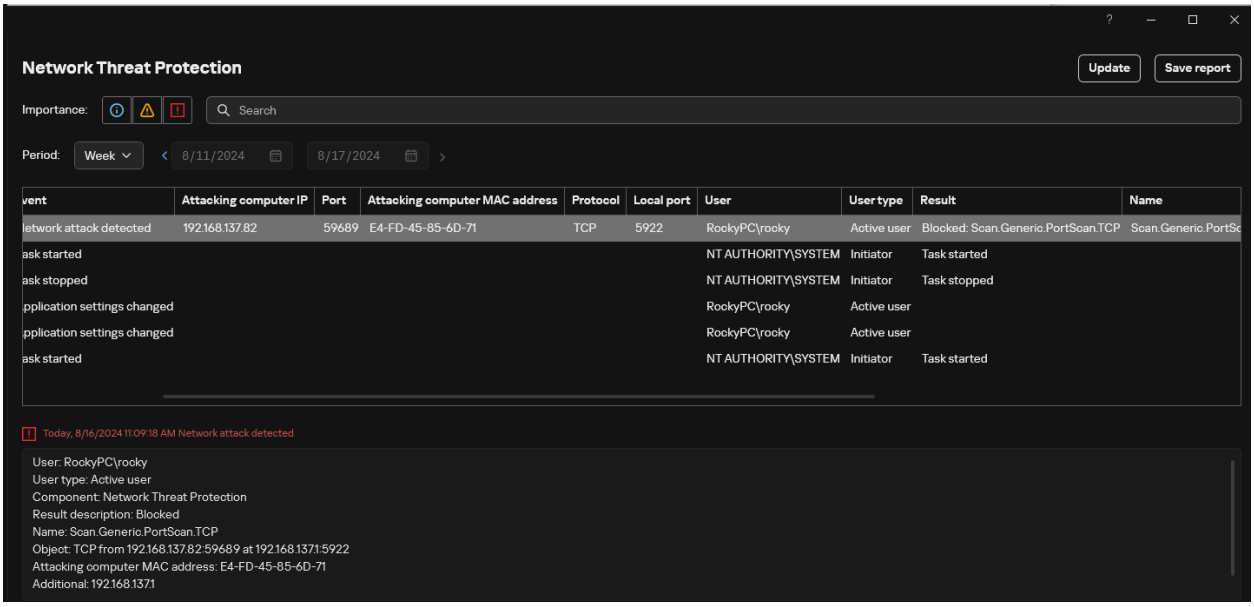

Kaspersky block scanning Traffic from Attack Machine

Benchmark 5 – Mail Threat Protection

The Mail Threat Protection component scans incoming and outgoing email messages for viruses and other threats. For EDRs with this feature, they continuously operate in the computer's memory and scan all messages sent or received via POP3, SMTP, IMAP, MAPI, and NNTP protocols. If no threat is detected in the email message, the message will be made available and/or processed.

Test case: Simulate a phishing attack where the attacker sends emails containing malware to mailboxes using Outlook with security solutions installed.

Kaspersky was able to scan email attachments, detect threats in files attachments and links in the contents of mail. However, Microsoft Defender did not take any action in this scenario.

Testcase	Microsoft Defender	Kaspersky
Malicious Email	Not Action	Blocked

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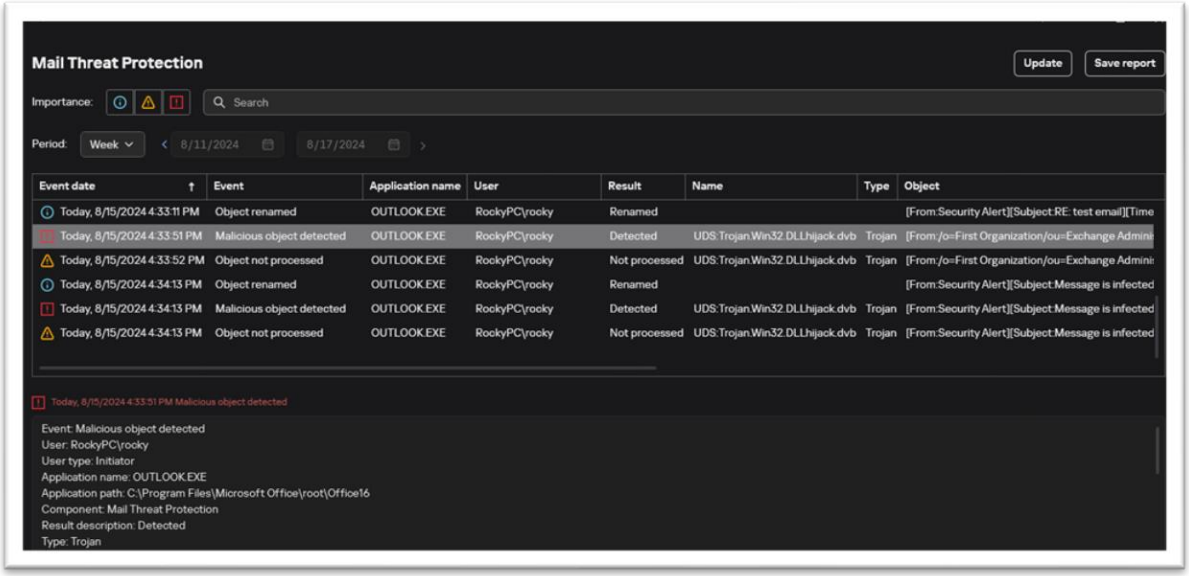
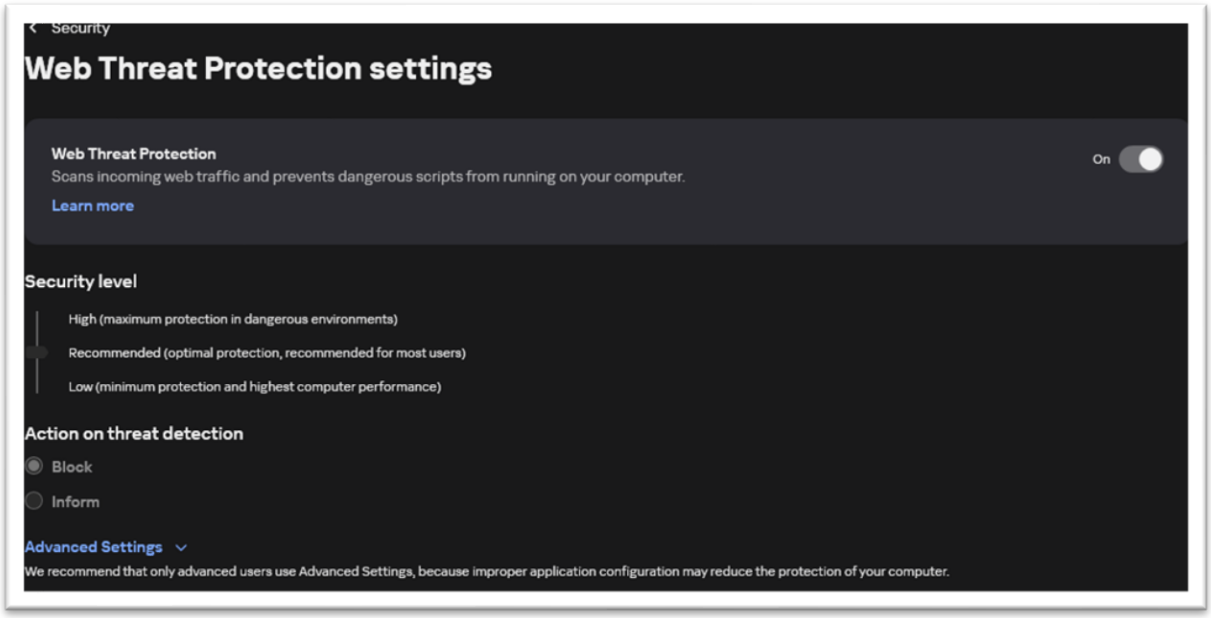


Kaspersky EDR Foundations blocked Email Malicious

Benchmark 6 – Web Threat Protection

Web Theat Protection helps reduce the attack surface of your devices from Internet-based events. It prevents employees from using any application to access dangerous domains that may host phishing scams, exploits, and other malicious content on the Internet.



Test case: Using the browser of your choice (Google Chrome), navigate to the suburand.com (malicious site) website test. Google Chrome has other security measures in place to protect from this vulnerability.

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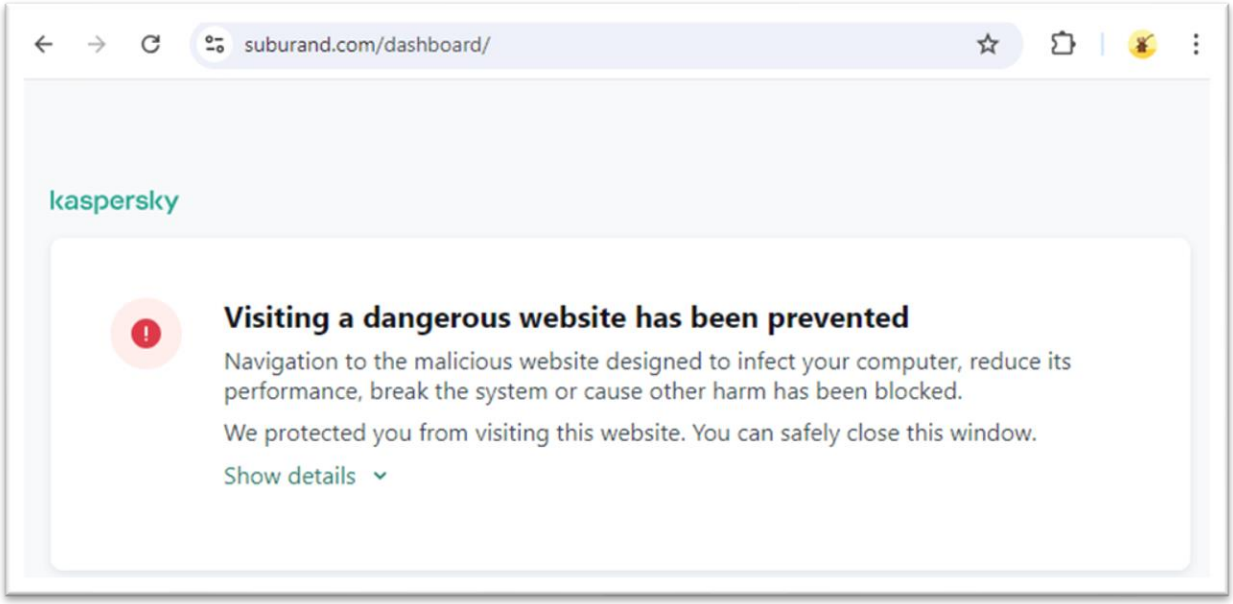
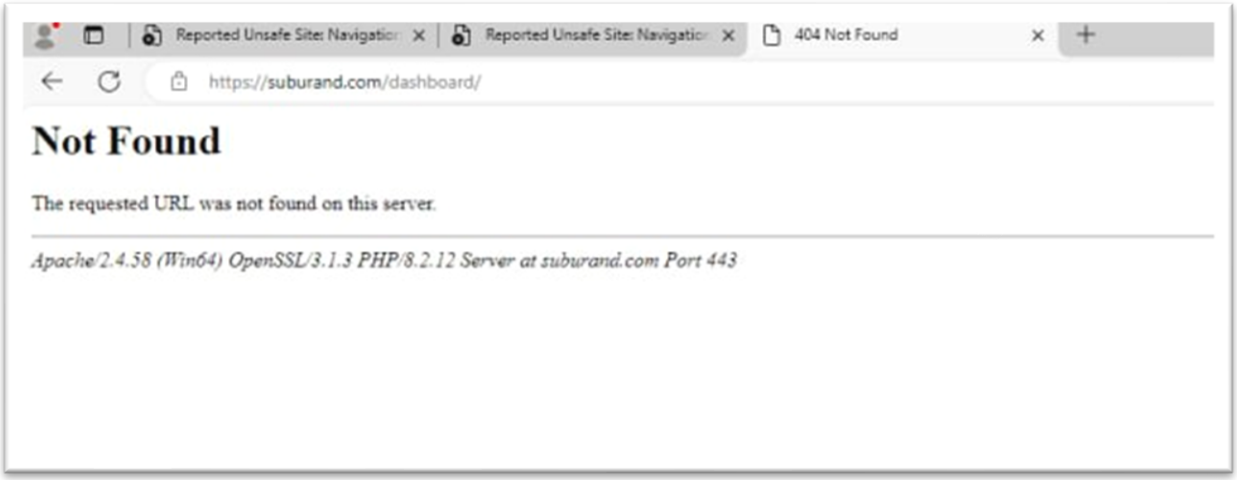
Kaspersky identifies malicious domains more accurately and more extensively than Microsoft Defender.

Testcase	Microsoft Defender for Endpoint	Kaspersky EDR Foundations
Malicious suburand.com website	Not Detected	Detected



Kaspersky EDR Foundations blocked suburand.com website



Microsoft Defender for Endpoint doesn't block suburand.com website

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Benchmark 7 – Quarantine Malicious Non-PE Files

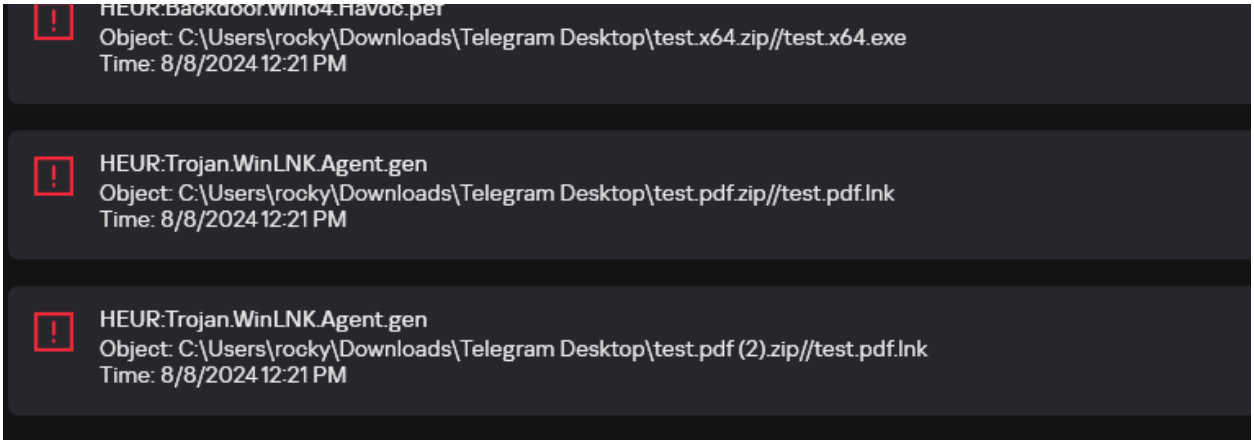
"Quarantine Malicious Non-PE Files" is a cybersecurity process in which non-Portable Executable (PE) malicious files are identified and isolated from the system to prevent spread or damage.

- Image files: .jpg, .png, .gif can be attacked by embedding malware in the metadata.
- Compressed files: .zip, .rar can contain malware in compressed form.

Test case: We drop malware embedded in a .zip compressed file onto the system and monitor the actions of both solutions.

Both solutions quarantine the .zip file. However, Kaspersky actively scans nearly all files within the .zip file and isolates unsafe files. *Some cases Microsoft Defender may only quarantine and alert after extraction.*

Testcase	Microsoft Defender for Endpoint	Kaspersky EDR Foundations
Malicious .zip file	Alerted/Blocked	Alerted/Blocked



Kaspersky scan and block .zip malicious file

Benchmark 8 – Device Control

Device Control ensures the security of sensitive data by restricting user access to devices installed on or connected to the computer, including:

- Data storage devices (hard drives, portable drives, tape drives, CD/DVD drives)
- Data transfer tools (modems, external network cards)
- Devices designed to convert data into hard copies (printers)
- Connection buses (also simply called "buses"), referring to interfaces for connecting devices to the computer (such as USB, FireWire, and Infrared)

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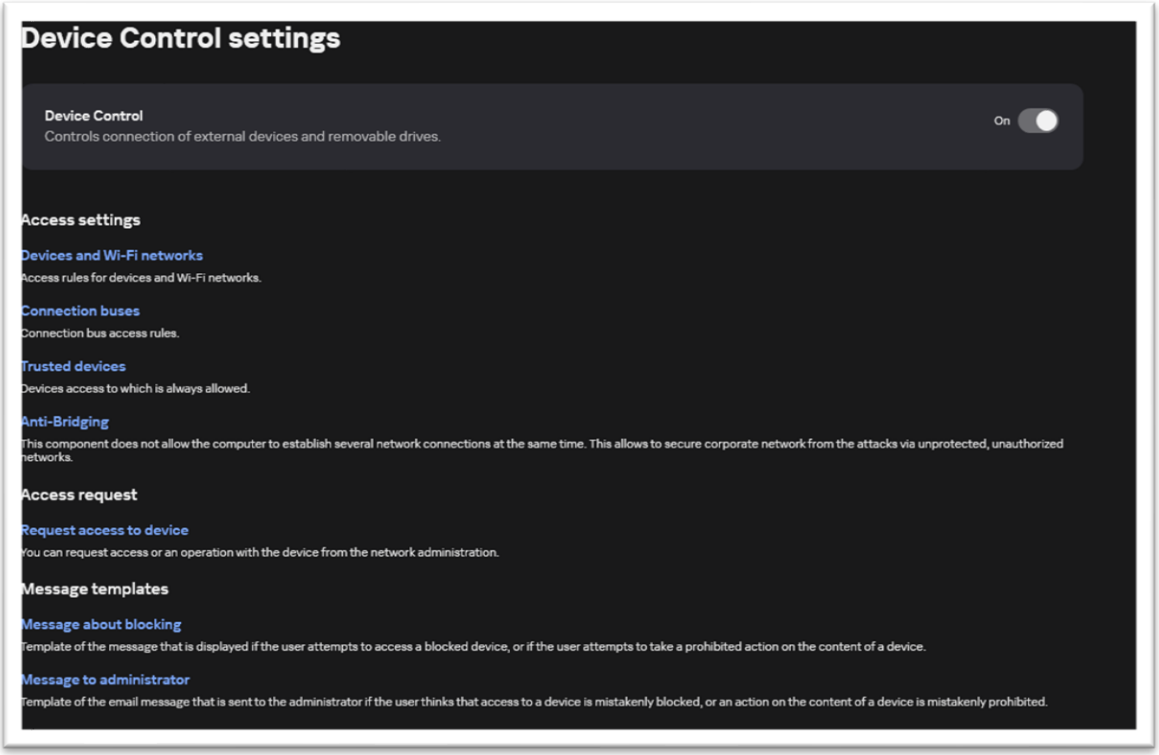

Device Control Setting

Test case: Explore and discover, experience the Device Control features on both solutions.

Both solutions support Device Control, but Microsoft Defender's Device Control management *requires the use of Microsoft Intune for support.*

	Defender	Kaspersky
Device Control	Support	Support
Management	Microsoft Intune	Admin Dashboard

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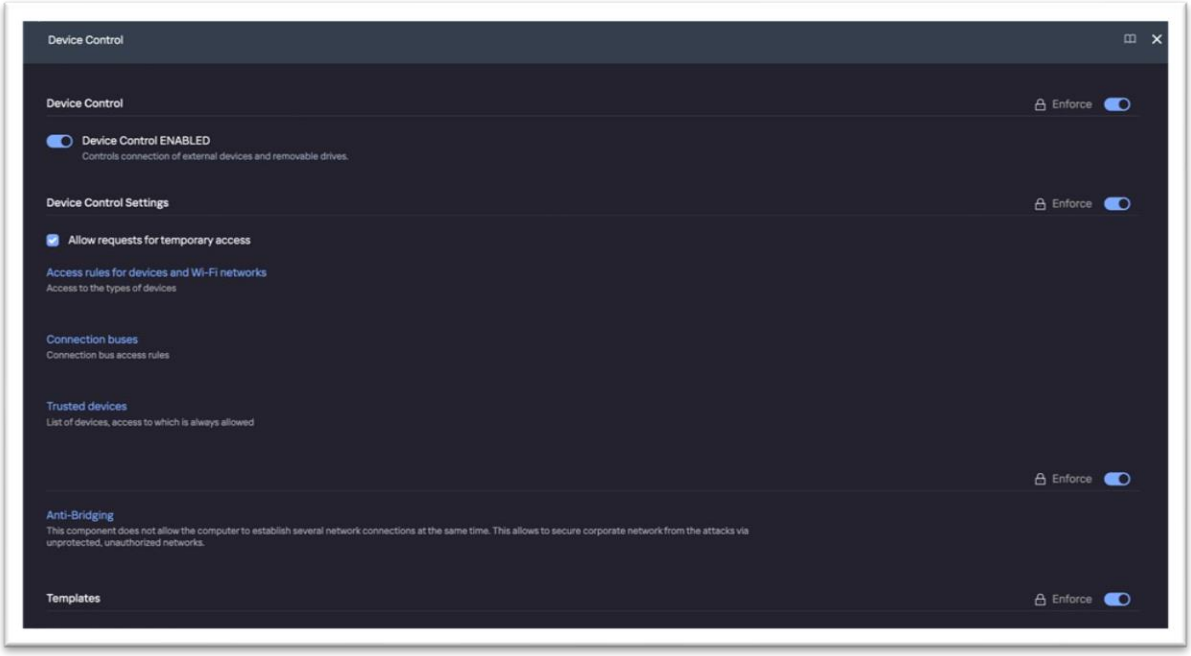
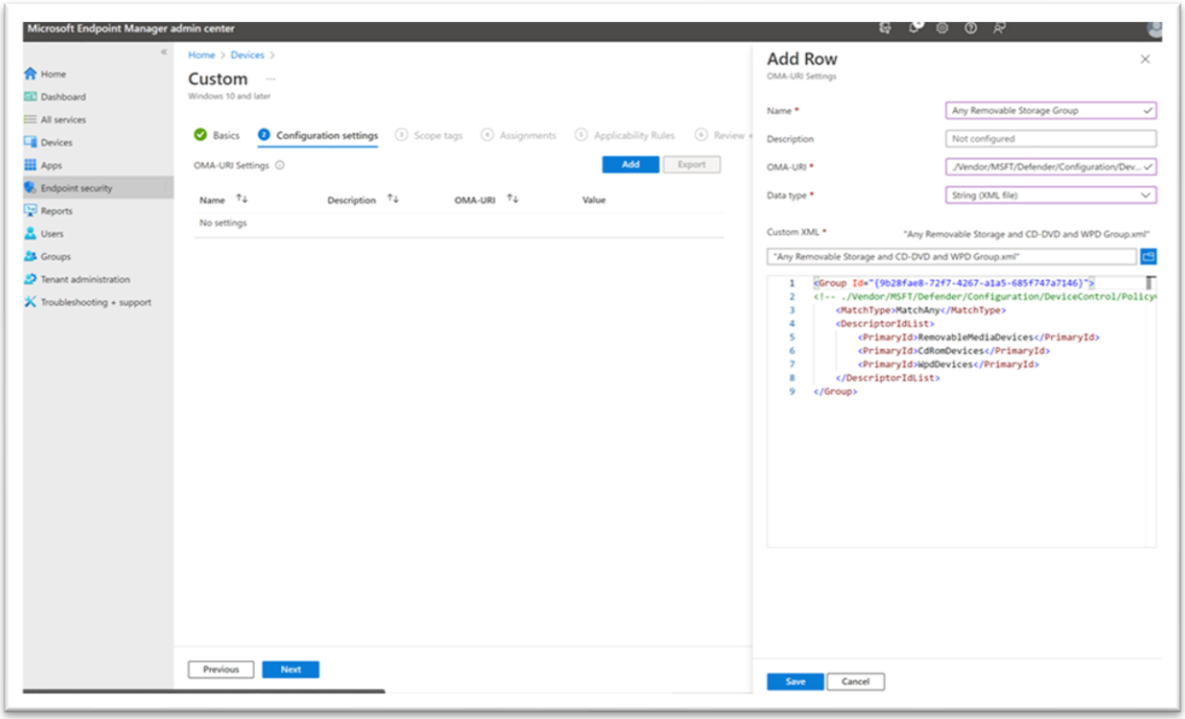


Kaspersky Device Control Management in Dashboard



Device Control Management in Microsoft Intune

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Benchmark 9 – Behavioral-Based Detection of Threats

Behavior-based threat detection is a method of identifying threats in cybersecurity by monitoring and analyzing the behavior of entities in the system (such as users, devices, software) to identify unusual or suspicious activities, rather than relying solely on signature-based detection.

This method combines various techniques for higher accuracy, such as using Machine Learning and Sandboxing for monitoring. Therefore, behavior-based threat detection is a key factor in determining whether an AV/EDR solution is effective.

Test case POC: We created a piece of malware that employs certain AV/EDR bypass techniques and dropped it onto computers with both solutions installed.

Kaspersky EDR Foundations detected the malware, while Microsoft Defender did not.

	Microsoft Defender	Kaspersky EDR Foundations
Output	Not Detected	Detected
Evidence	Video POC	Video POC

Benchmark 10 – Signature-based Detection of Threats

Signature-based Detection of Threats is a cybersecurity method that involves comparing files, network traffic, or system activities with a database containing signature patterns of known threats, such as malware, viruses, or cyberattacks.

Test case: The team used several new signatures discovered from the community to test which solution, Kaspersky or Microsoft Defender, identifies the signatures earlier and more extensively.

Date (UTC)	SHA256 hash	Type	Signature	Tags	Reporter	DL
2024-08-16 09:39	2890e9db681b27690786...	exe		exe	SecuriteInfoCom	
2024-08-16 09:36	85b919ece43dc0c15de0...	exe		exe scr	abuse_ch	
2024-08-16 09:36	d7b0eafed04c1ae1bdf98...	exe		exe scr	abuse_ch	
2024-08-16 09:34	a5cb30c30d046a1a6d74...	exe		exe scr	abuse_ch	
2024-08-16 09:32	71512deeed8878e88803...	exe		exe scr	abuse_ch	
2024-08-16 09:26	7bcdc2e607abc65ef93af...	exe		exe	UkyKnight	
2024-08-16 09:26	327ee1eccb682dfab794e...	exe		exe	UkyKnight	
2024-08-16 09:25	998b6a7ad1579c31d13a...	exe		exe	UkyKnight	
2024-08-16 09:25	53f4e080c306971a2dba4...	dll		dll	UkyKnight	
2024-08-16 09:17	584cc955627e1a6180c4e...	exe	SnakeKeylogger	exe SnakeKeylogger	threatcat_ch	
2024-08-16 09:17	ca23e4b49d8618c8b4b0...	exe		exe	abuse_ch	
2024-08-16 09:17	9e7487c7199e53a48d1d...	exe		exe	abuse_ch	
2024-08-16 09:17	b7823204f17c757d3724...	exe	GuLoader	exe GuLoader	abuse_ch	
2024-08-16 09:17	76352129fb5c6724d764...	exe	GuLoader	exe GuLoader	abuse_ch	
2024-08-16 09:16	1ac6298f7cda650483d81...	exe	GuLoader	exe GuLoader	abuse_ch	

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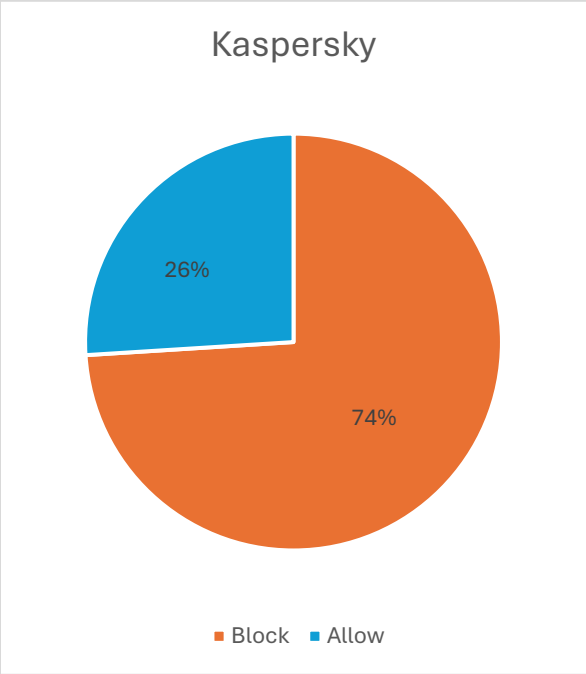
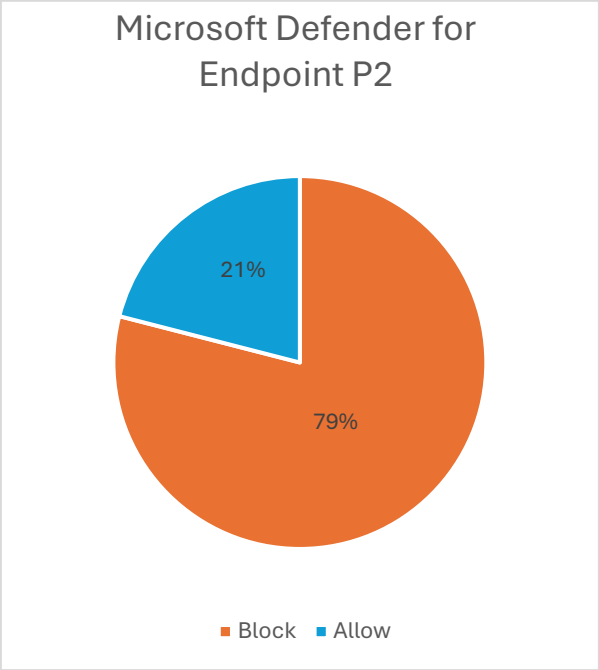
Microsoft Defender identifies threats based on signatures better than Kaspersky. Since Microsoft Defender is built into Windows, it is understandable that it collects samples and signatures more effectively compared to Kaspersky.

Testcase	Defender	Kaspersky
Detected Signature	29/34	25/34





Benchmark 11 – Real-Time Monitoring System Activity

Real-Time Monitoring System Activity is the process of continuously observing and tracking activities occurring on a computer system or network as they happen. The goal of real-time monitoring is to detect, record, and respond immediately to events or behaviors that may pose risks, including cybersecurity threats, system errors, or unwanted activities.

Test case: Explore and discover, experience Real-Time Monitoring System Activity features on both solutions.

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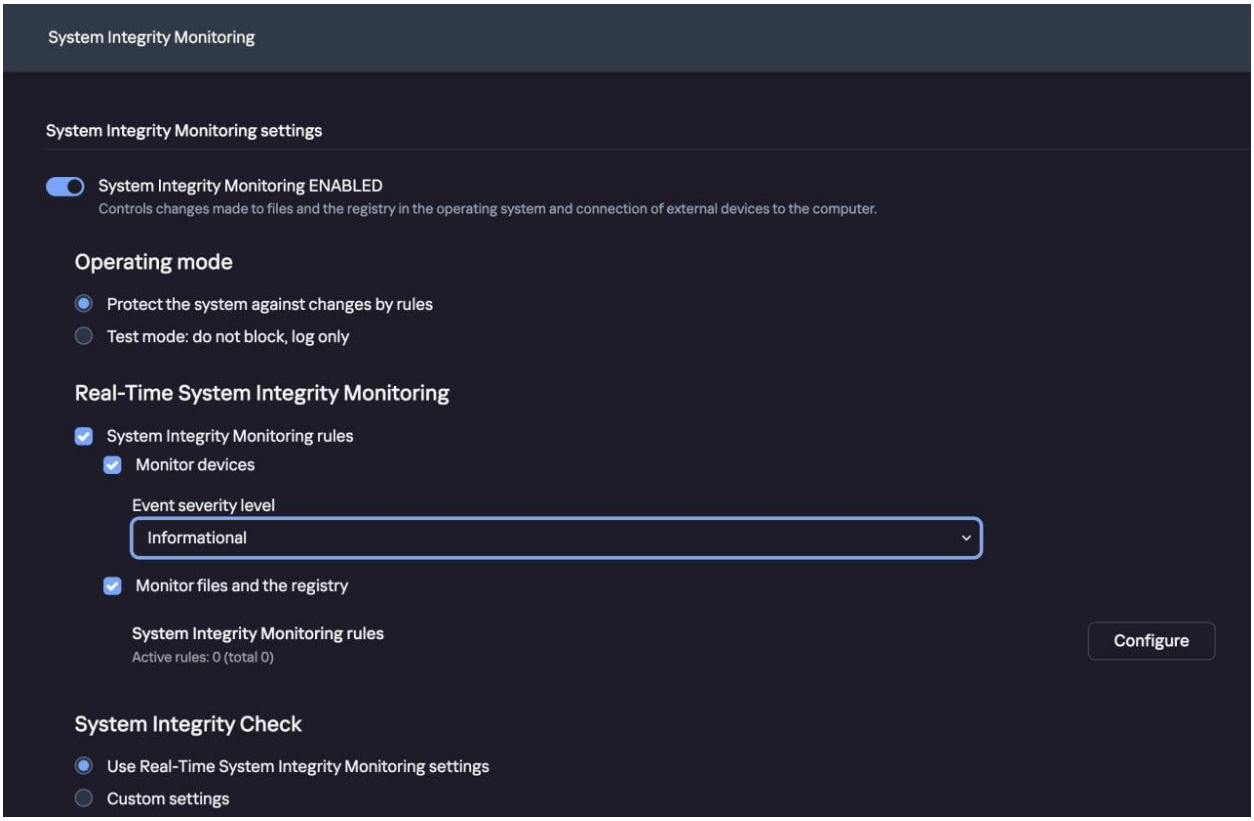

Kaspersky supports this feature and applies it easily, but for Defender, we could not find this feature.

Microsoft Defender for Endpoint Plan 2	Kaspersky EDR Foundations
• N/A	• Support

Benchmark 12 – Sandbox Analyzer for Dynamic Analysis

Sandbox Analyzer for Dynamic Analysis is a tool or special environment used in cybersecurity to analyze the behavior of unknown software or code in real-time. The "sandbox" here refers to an isolated virtual environment where suspicious files or applications can be executed without affecting the actual system.

Test case: Explore and discover Sandbox features on both solutions.

Microsoft Defender	Kaspersky EDR Foundations
Included in deep file analysis (Plan2) without local sandbox. Therefore, threat detection so slowly. https://learn.microsoft.com/en-us/defender-endpoint/sandbox-mdav	Locally available, Kaspersky Sandbox technology to scan only executable files (e.g. EXE) and Microsoft Office files (DOC, DOCX, XLS, PPT and others), so threat detection is fast. https://support.kaspersky.com/KESWin/11.2.0/en-us/190288.htm

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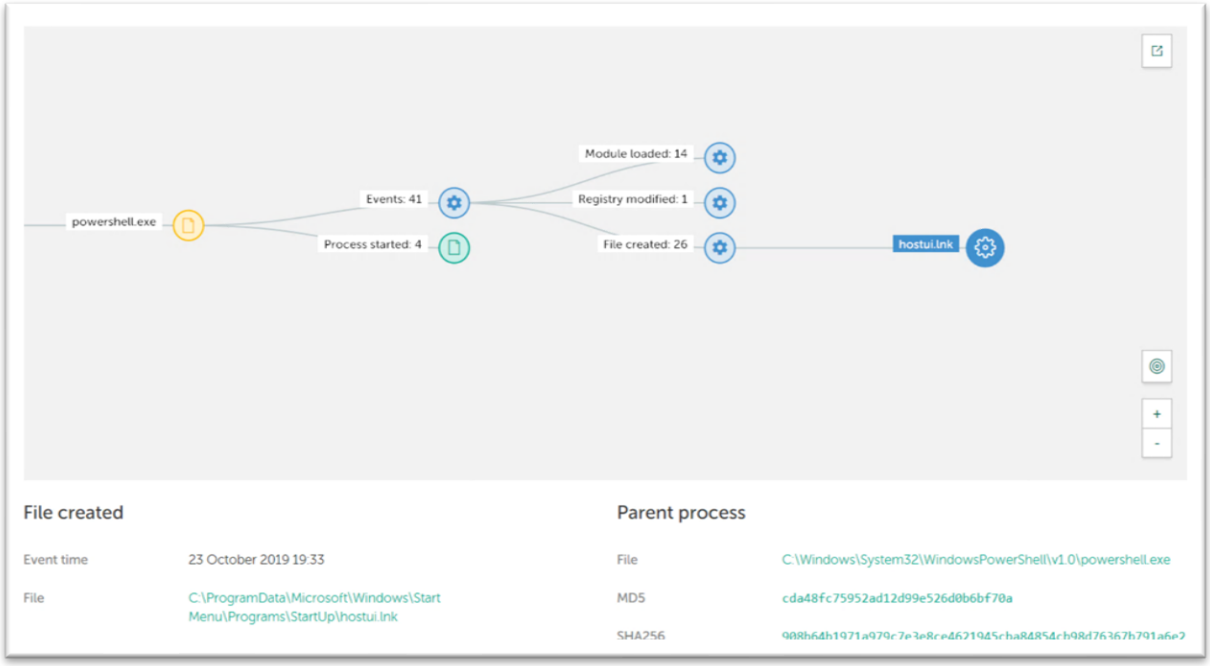
Benchmark 13 – Unusual Files and Registry Entries

Unusual Files and Registry Entries Detection is a method in cybersecurity used to identify files or registry entries in the system that exhibit unusual or abnormal characteristics compared to the system's normal activity. This is an important part of security solutions, helping to detect early signs of malicious activity or intrusions.

Test case: Perform the attack using powershell.exe to create the hosting.lnk file in the Startup folder.

Both solutions can detect the threat from this behavior.

Microsoft Defender for Endpoint Plan 2	Kaspersky EDR Foundations
• Detected	• Detected



Kaspersky detect the attack scenario

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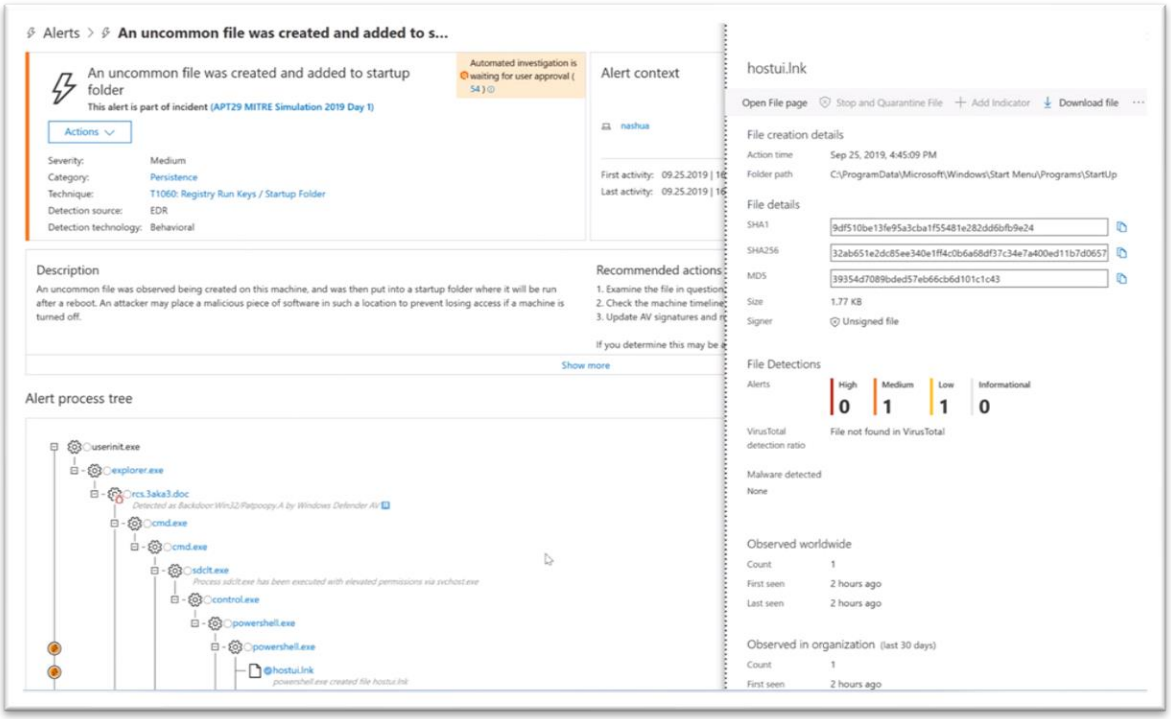

Microsoft Defender detect the attack scenario

Benchmark 14 – Fileless Malware

Fileless Malware is a type of malware that does not use files like traditional malware. Instead, it exploits legitimate applications or processes running on the system to carry out malicious actions. This makes fileless malware difficult to detect by traditional antivirus software, as it does not leave specific file traces on the disk.

Test case: Assume the attack scenario uses cmd.exe spawning from the rcs.3aka3.doc "fileless" process.

Both solutions were able to identify the threat from this behavior.

Microsoft Defender for Endpoint Plan 2	Kaspersky EDR Foundations
• Detected	• Detected

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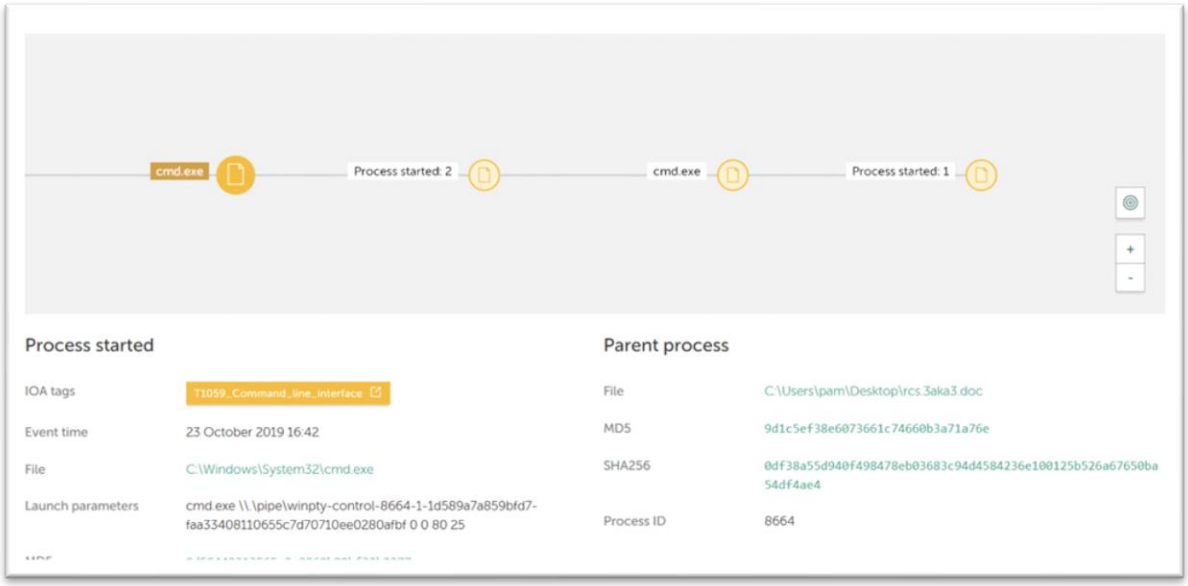
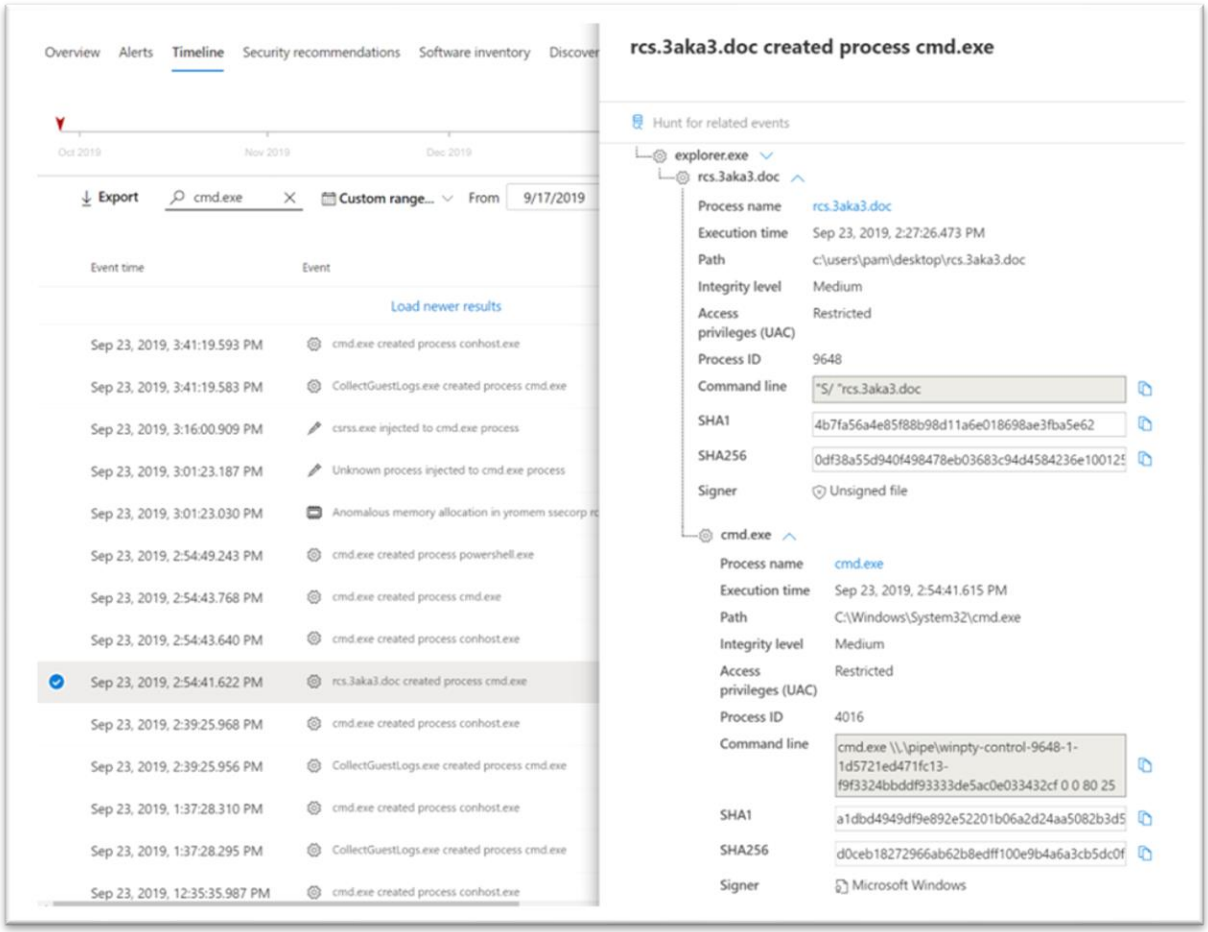


Kaspersky detected threats from Fileless malware



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Microsoft Defender detected threats from Fileless malware

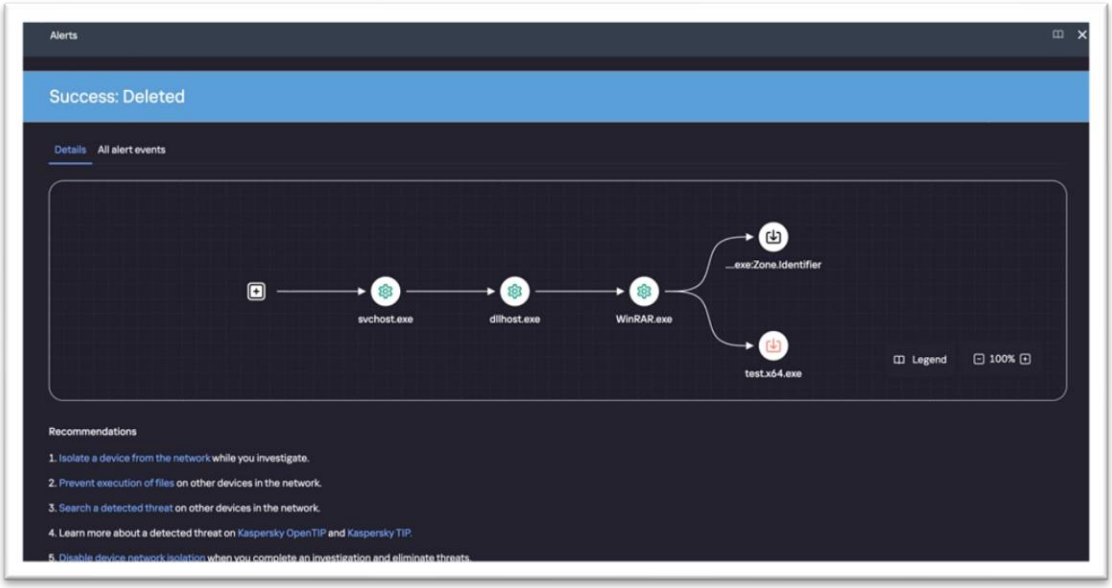
Benchmark 15 – Root Cause Analysis

Root Cause Analysis allows you to detect and eliminate advanced attacks at their source by performing root cause analysis using a visualized threat development chain diagram and diving into details for further examination.

Test case: Explore and discover, experience the firewall features on both solutions.

Both solutions support this feature and are user-friendly, making it easy for administrators to operate the system. However, each product has its own strengths and advantages as follows:

Microsoft Defender for Endpoint Plan 2	Kaspersky EDR Foundations
• • Automation and integration of RCA with other Microsoft security solutions, providing a seamless experience for investigation and troubleshooting. • Integration with Threat Analytics, providing insights into known threats and corresponding mitigation measures.	• Detailed and accurate but *may lack strong automation and integration*. • Focusing on providing insights into malicious entities, dangerous actions, and affected processes.



Root Cause Analysis dashboard from Kaspersky

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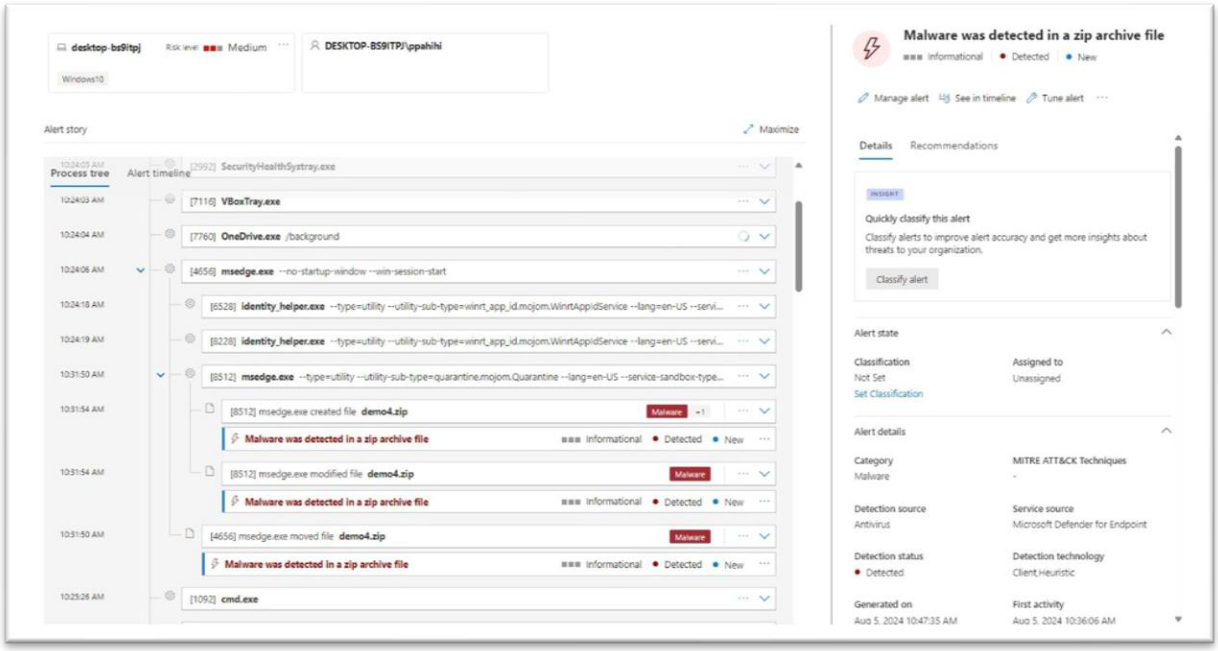

Root Cause Analysis dashboard from Microsoft Defender

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